

NANYANG TECHNOLOGICAL UNIVERSITY  
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 1

MH1100 Calculus I

Tutorial 1, Week 2

**Your tutor will aim to discuss: Problem 1, 2, 5, 10, and 13**

**Problem 1** What is the domain of the function

$$f(x) = \frac{1}{\sqrt{3} - 2 \cos x}.$$

[Solution:] This expression makes sense when  $\sqrt{3} - 2 \cos x \neq 0$ .  $\sqrt{3} - 2 \cos x$  is zero when  $\cos x = \sqrt{3}/2$ .  $\cos x$  is a periodic function, which has a period of  $2\pi$ . In the interval  $[-\pi, \pi]$ , when  $x = \pm\pi/6$ ,  $\cos x = \sqrt{3}/2$ . Hence, we get that  $\cos x = \sqrt{3}/2$  when  $x = 2n\pi \pm \pi/6$ , and the domain of the function  $f$  is

$$\left\{ x \in \mathbb{R} \mid x \neq (2n \pm \frac{1}{6})\pi, n = 0, \pm 1, \pm 2, \dots \right\}$$

**Problem 2** The function whose value at any number  $x$  is the *greatest integer less than or equal to  $x$*  is called the **greatest integer function** or the **integer floor function**. It is denoted  $\lfloor x \rfloor$ . Draw the graph of the integer floor function.

[Solution:] The graph of the greatest integer function  $\lfloor x \rfloor$  is

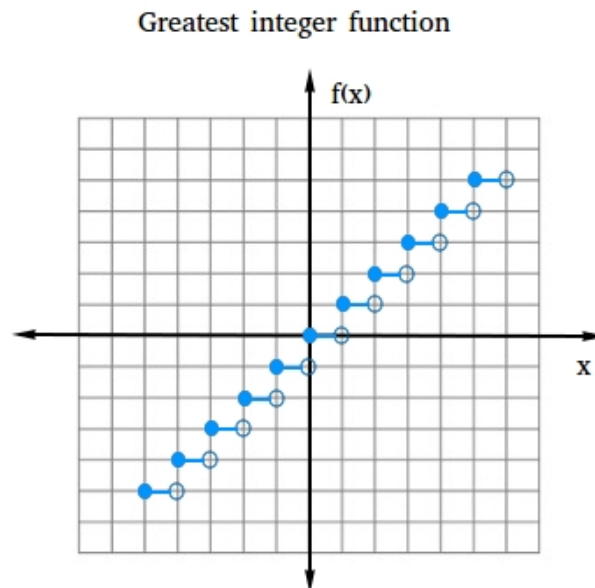


FIGURE 1. The graph of  $\lfloor x \rfloor$ .

**Problem 3** The function whose value at any number  $x$  is the *smallest integer greater than or equal to  $x$*  is called the **least integer function** or the **integer ceiling function**. It is denoted  $\lceil x \rceil$ . Draw the graph of the integer ceiling function.

[Solution:] The graph of the greatest integer function  $\lfloor x \rfloor$  is

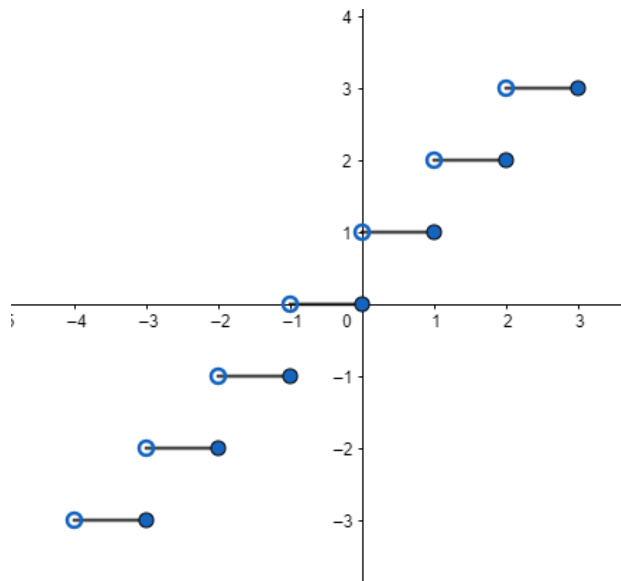


FIGURE 2. The graph of  $\lceil x \rceil$ .

**Problem 4** Show that if  $f$  is both even and odd, then  $f(x) = 0$  for every  $x$  in the domain of  $f$ .

[Solution:] If  $f(x)$  is an even function,

$$f(x) = f(-x).$$

If  $f(x)$  is an odd function,

$$f(x) = -f(-x).$$

Summing up the above two equations gives that

$$2f(x) = f(-x) - f(-x) = 0.$$

Thus,  $f(x) = 0$  for every  $x$  in its domain. This completes the proof.

**Problem 5** Show that the function  $f(x)$  below is odd,

$$f(x) = \ln \left( x + \sqrt{1 + x^2} \right).$$

[Solution:]

$$\begin{aligned}
 f(-x) &= \ln(-x + \sqrt{1 + (-x)^2}) = \ln(-x + \sqrt{1 + x^2}) \\
 &= \ln\left[\left(-x + \sqrt{1 + x^2}\right) \frac{x + \sqrt{1 + x^2}}{x + \sqrt{1 + x^2}}\right] \\
 &= \ln\left[\frac{1}{x + \sqrt{1 + x^2}}\right] \\
 &= -\ln(x + \sqrt{1 + x^2}) \\
 &= -f(x)
 \end{aligned}$$

That is  $f(-x) = -f(x)$ .  $f$  is odd. This completes the proof.

**Problem 6** Prove that  $f(x) = \frac{x}{1-x}$  is increasing on the interval  $(-\infty, 1)$ .

[Solution:] Choosing any two points  $x_1 < x_2$  in the interval  $(-\infty, 1)$ , we have

$$\begin{aligned}
 f(x_1) - f(x_2) &= \frac{x_1}{1-x_1} - \frac{x_2}{1-x_2} \\
 &= \frac{x_1(1-x_2) - (1-x_1)x_2}{(1-x_1)(1-x_2)} \\
 &= \frac{x_1 - x_2}{(1-x_1)(1-x_2)} \\
 &< 0.
 \end{aligned}$$

That means for every pair of  $x_1$  and  $x_2$  in  $(-\infty, 1)$ , if  $x_1 < x_2$ , then  $f(x_1) < f(x_2)$ . Thus,  $f(x)$  is increasing in  $(-\infty, 1)$ .

**Problem 7** Find the domain of  $f(x) = \frac{|x|}{x^2}$ , sketch the function, and find the interval where  $f$  is an increasing function.

[Solution:] First, notice that this expression makes sense when evaluated at any  $x$ , except  $x = 0$ . Thus, the domain of the function  $f(x)$  is  $\mathbb{R} \setminus \{0\}$ .

To plot  $f$ , we first work out the rules which determine  $f$  in the different regions:

$$\begin{aligned}
 f(x) = \frac{|x|}{x^2} &= \begin{cases} \frac{(-x)}{x^2}, & x < 0 \\ \frac{(+x)}{x^2}, & x > 0 \end{cases} \\
 &= \begin{cases} -\frac{1}{x}, & x < 0 \\ \frac{1}{x}, & x > 0 \end{cases}
 \end{aligned}$$

From the graph on the next page we know that  $f$  is a decreasing function on the interval  $(0, \infty)$ . For any two points  $x_1$  and  $x_2$  from  $(0, \infty)$ , we can easily show that  $f(x_1) > f(x_2)$ . Likewise, for any two points  $x_1$  and  $x_2$  from  $(-\infty, 0)$ , we can also show that  $f(x_1) < f(x_2)$ . So we can say that  $f$  is a decreasing function on the interval  $(0, \infty)$ , and it is an increasing

function on the interval  $(-\infty, 0)$ . But  $f(x)$  is neither a decreasing function nor an increasing function on its domain  $\mathbb{R} \setminus \{0\}$ .

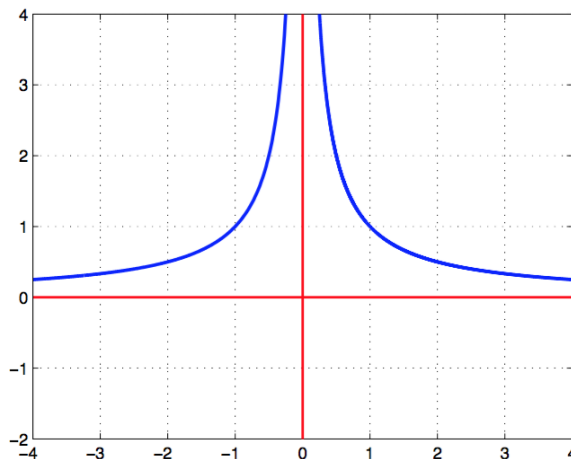


FIGURE 3. The graph of  $f(x) = \frac{|x|}{x^2}$ .

**Problem 8** Starting with the graph of  $y = 1/x$ , use geometric transformations to plot the graph of the function

$$f(x) = \frac{3}{x-1}.$$

[Solution:] Let  $g(x) = \frac{1}{x}$ . Note that

$$f(x) = 3 \cdot g(x-1).$$

So the graph we want is obtained from the standard graph  $g(x) = \frac{1}{x}$  by shifting it one unit to the right then stretching it along the  $y$ -direction by a factor of 3.

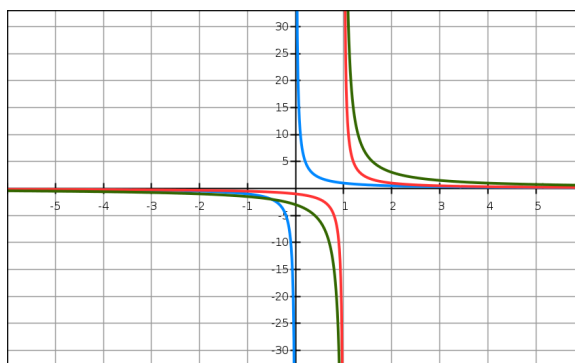


FIGURE 4. From  $y = 1/x$  (blue) to  $y = 1/(x-1)$  (red) and to  $y = 3/(x-1)$  (green).

**Problem 9** Given  $f(x) = x^3 + 2x^2$  and  $g(x) = 3x^2 - 1$ , find (a)  $f + g$ , (b)  $f - g$ , (c)  $fg$ , (d)  $f/g$  and state their domains.

[Solution:]

The domain of  $f(x)$  is  $A = (-\infty, \infty)$ . The domain of  $g(x)$  is also  $B = (-\infty, \infty)$ .

(a)  $(f+g)(x) = f(x) + g(x) = (x^3 + 2x^2) + (3x^2 - 1) = x^3 + 5x^2 - 1$ . The domain of the sum function is  $A \cap B = (-\infty, \infty)$ .

(b)  $(f-g)(x) = f(x) - g(x) = (x^3 + 2x^2) - (3x^2 - 1) = x^3 - x^2 + 1$ . The domain of the difference function is  $A \cap B = (-\infty, \infty)$ .

(c)  $(fg)(x) = f(x)g(x) = (x^3 + 2x^2)(3x^2 - 1) = 3x^5 + 6x^4 - x^3 - 2x^2$ . The domain of the product function is  $A \cap B = (-\infty, \infty)$ .

(d)

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^3 + 2x^2}{3x^2 - 1}.$$

The domain of the quotient function is the intersection of the domain of  $f$  and the domain of  $g$  but excluding the point or points on which  $g(x) = 0$ . We find that when  $x = \pm\sqrt{3}/3$ ,  $g(x) = 0$ . So from the result of (c) we get that the domain of  $f/g$  is  $\mathbb{R} \setminus \{-\sqrt{3}/3, \sqrt{3}/3\}$ .

**Problem 10** If  $f(x) = \sqrt{x+1}$  and  $g(x) = 4x - 3$ , find the functions  $f \circ g$ ,  $g \circ f$ ,  $f \circ f$ , and  $g \circ g$ , and find their domains. [Solution:]

- The function  $f \circ g$  is given by the formula  $(f \circ g)(x) = \sqrt{(4x-3)+1} = \sqrt{4x-2}$ . Because  $g$  is defined for every  $x$ , and  $f$  is defined on the interval  $[-1, \infty)$ , the domain of  $f \circ g$  is  $\{x \in \mathbb{R} | g(x) \geq -1\}$ . That is  $\{x \in \mathbb{R} | x \geq 1/2\}$ .
- The function  $g \circ f$  is given by the formula  $(g \circ f)(x) = 4\sqrt{x+1} - 3$ .  $g$  is defined on the whole  $x$ -axis. So, the domain of  $g \circ f$  is precisely the domain of  $f(x)$  as  $[-1, \infty)$ .
- The function  $f \circ f$  is given by the formula  $(f \circ f)(x) = \sqrt{\sqrt{x+1}+1}$ . It is obvious that the domain of  $f \circ f$  is  $[-1, \infty)$ .
- The function  $g \circ g$  is given by the formula  $(g \circ g)(x) = 4(4x-3) - 3 = 16x - 15$ . The domain is obviously the whole set of real numbers  $\mathbb{R}$ .

### Problem 11

- Suppose  $g(x)$  is an even function and we set  $h = f \circ g$ , where  $f$  is some other function. Is  $h(x)$  even?
- Suppose  $g(x)$  is an odd function and we set  $h = f \circ g$ , where  $f$  is some other function. Is  $h(x)$  always odd?
- Suppose  $g(x)$  is an odd function and we set  $h = f \circ g$ , where  $f$  is odd. What about  $h(x)$ ?
- Suppose  $g(x)$  is an odd function and we set  $h = f \circ g$ , where  $f$  is even. What about  $h(x)$ ?

[Solution:]

- Since  $g(x)$  is an even function, we have  $g(-x) = g(x)$  for every  $x$ . Let's test the same property for the function  $h(x)$

$$h(-x) = (f \circ g)(-x) = f(g(-x)) = f(g(x)) = (f \circ g)(x) = h(x).$$

The third equality is true because  $g(x)$  is even. Thus we can conclude that  $f \circ g$  is also even.

- That  $g(x)$  is an odd function indicates that  $g(-x) = -g(x)$  for every  $x$ . Let's test the same property for the function  $h(x)$

$$h(-x) = (f \circ g)(-x) = f(g(-x)) = f(-g(x)).$$

The third equality is true because  $g(x)$  is odd. The minus sign in  $f(-g(x))$  cannot come out or disappear given that  $f$  is only some other function. That means  $h(x)$  is not always odd.

- From (b) we know that if  $f$  is odd, then  $h(-x) = f(-g(x)) = -f(g(x)) = -h(x)$ . Thus  $h(x)$  is odd.
- From (b) we know that if  $f$  is even, then  $h(-x) = f(-g(x)) = f(g(x)) = h(x)$ . Thus  $h(x)$  is even.

**Problem 12** Let  $f(x) = ax + b$  and  $g(x) = cx + d$ . What condition must be satisfied by the constants  $a, b, c, d$  in order that  $(f \circ g)(x) = (g \circ f)(x)$  for every value of  $x$ ?

[Solution:] The composite functions are  $(f \circ g)(x) = a(cx + d) + b$  and  $(g \circ f)(x) = c(ax + b) + d$ . Since  $(f \circ g)(x) = (g \circ f)(x)$ , we have  $a(cx + d) + b = c(ax + b) + d$ . So,  $ad + b = bc + d$ .

**Problem 13** For any given real numbers  $a$  and  $b$ , prove that  $|a| - |b| \leq |a \pm b| \leq |a| + |b|$ .

[Solution:] We know that

$$-|a| \leq a \leq |a|, \quad -|b| \leq b \leq |b|.$$

Summing up the two expressions above gives that

$$-(|a| + |b|) \leq a + b \leq |a| + |b|.$$

Thus,

$$|a + b| \leq |a| + |b|.$$

If  $b$  is replaced by  $-b$ , we further have

$$|a - b| \leq |a| + |-b| = |a| + |b|.$$

We have proved that  $|a \pm b| \leq |a| + |b|$ . To prove  $|a| - |b| \leq |a \pm b|$ , we have

$$|a| = |a - b + b| \leq |a - b| + |b|,$$

which suggests that

$$|a| - |b| \leq |a - b|.$$

Using  $-b$  to replace  $b$  gives

$$|a| - |-b| \leq |a + b|.$$

So,  $|a| - |b| \leq |a \pm b|$ . We have proved  $|a| - |b| \leq |a \pm b| \leq |a| + |b|$ .

**Problem 14** A spherical balloon is being inflated and the radius of the balloon is increasing at a rate of 2 cm/s.

- (a) Express the radius  $r$  of the balloon as a function of the time  $t$  (in seconds).

- (b) If  $V$  is the volume of the balloon as a function of the radius, find  $V \circ r$  and interpret it.

[Solution:]

- (a) The radius  $r$  of the balloon can be expressed as

$$r(t) = 2t + c.$$

Here  $c$  is a constant, the initial radius of the balloon.

- (b) If  $V$  is the volume of the balloon as a function of the radius, then

$$V(r) = \frac{4}{3}\pi r^3.$$

Thus,  $V \circ r$  can be written as

$$V \circ r(t) = \frac{4}{3}\pi(2t + c)^3.$$

$V \circ r$  describes how the volume of the balloon changes through the time  $t$ .